

SUN[™] STORAGE 7110, 7210, AND 7410 UNIFIED STORAGE SYSTEMS

Architected for Open, Simple, and Scalable Enterprise Storage

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Executive Summary

Near exponential data growth is changing the rules for organizations everywhere. Server and desktop virtualization, databases, Web 2.0 applications and high-performance computing (HPC) are all contributing to new data management challenges, even as these applications change the way that organizations use data to fulfill their missions and impact their bottom lines. Streaming and transactional data in particular are driving new requirements for storage infrastructure.

Complicating matters, these challenges are occurring in the context of a highly-competitive global marketplace where getting to data to market quickly with predictable costs can make all of the difference. Like computational infrastructure, data storage infrastructure must be agile in order to scale for unpredictable growth spikes in workloads and changing business strategies. With volatile global economic conditions, no organization can afford to ignore costs, even as they plan for future growth and deploy storage infrastructure that can continue to perform at peak levels.

Unfortunately, most of today's storage solutions remain proprietary, complex, and expensive, with appliance vendors seeking proprietary lock-ins and lucrative software licensing. In this demanding environment, special-purpose appliances have hit hard limitations in terms of performance and scalability. Limits for power, cooling and real estate have also become very real constraints, and energy costs are rising while IT budgets remain static. New constraints for power and cooling are also driving many to rethink the way that they deploy both computational and storage infrastructure.

Building on over 25 years of open system innovation and leadership, Sun™ Storage 7000 Series Unified Storage Systems provide a refreshing and much-needed change. As the world's first open storage appliances, these systems are a radically simple way to use storage at a fraction of the cost of traditional systems. Based on industry-standard components, Sun Storage 7000 Management software, and supported by a passionate community of developers and ISVs, Sun Storage 7000 Unified Storage Systems are up to ten times faster to install than competing systems, and require no special training to configure and use.

These systems also feature the industry's most comprehensive DTrace Analytics environment, including innovative new tools to help isolate and resolve issues before they impact the business. Sun Storage 7000 Unified Storage Systems are the only storage systems with Hybrid Storage Pools that automatically optimize performance while lowering power and cooling requirements. Perhaps most importantly, the Sun Storage 7000 family delivers up to twice the performance and capacity of competing offerings, breaking down performance and scalability barriers and delivering breakthrough return on investment (ROI). This document details the key architectural aspects of Sun Storage 7000 Unified Storage Systems.

Chapter 1

Changing Storage Economics

Most organizations today are digitized, online, and managing digital assets that are continuing to expand. Applications that rely upon file-centric storage infrastructure in particular are accounting for increasingly greater percentages of corporate data. This unstructured data increasingly must be managed in a cost-effective fashion if these organizations are to continue to grow, scale, and operate into the future.

Unceasing Demand for Storage

Organizations across a broad swath of industries, and running a variety of applications, are struggling to keep pace with rapid storage growth. These firms are looking for a radically easier and faster way to manage storage with a substantially better return on investment (ROI). A variety of applications are driving an appetite for new storage.

- *Web 2.0 applications* run the gamut from WeBlogs (Blogs), wikis, podcasts and social networking applications to Web serving, search engines, and video streaming. New applications arrive daily, changing the ways that users create, manipulate, and store data. No matter what the operating system platform, storage infrastructure must be flexible and scalable. At the same time, operators need deep insight into data management telemetry.
- *High-performance Computing (HPC) applications* in markets from manufacturing to energy to scientific pursuit are now creating simply unprecedented amounts of data. Increasingly large computational clusters of teraflop and petaflop scale are generating more data than ever before. Energy companies need storage for highly-parallel data streams and seismic data processing. Scientific datacenters need storage for pre-processing and post-processing of data in order to promote understanding of complex problems.
- *Server virtualization* is increasingly popular as organizations consolidate larger numbers of slower systems onto fewer, faster servers. Along with computational horsepower, servers used for virtualization need fast access to reliable storage. Traditionally, combining compute and storage workloads was extremely complex. Virtualization simplifies this process with a separate instance of combined workloads that can be warehoused into a single server platform.
- *Databases* and related applications such as data warehousing and business intelligence are also driving large demand for flexible data storage infrastructure. Storage infrastructure must be reliable to prevent business interruptions and improve business productivity. Most importantly, storage infrastructure must effectively support rapid business growth and unpredictable business conditions.

Open Unified Storage Arrives

As the productivity and viability of diverse organizations has become every more reliant on the ability to store, manage, and share information, network attached storage (NAS) appliances have become increasingly popular. These appliances attach directly to the network, and contain an operating system that has been customized and optimized to store and share files. The ease of use presented by NAS appliances, along with their dedicated purpose to store, share, and manage files has made them very popular for file-level services.

The concept of unified storage systems have become even more popular, providing a solution that not only supports NAS protocols, but but also provides storage area network (SAN) connectivity through iSCSI or Fibre Channel protocols. Many organizations prefer a unified approach since it provides options in terms of the protocols that can be run to access data. As a result, more diverse application data can be moved to a unified storage appliance where it can be managed effectively.

Unfortunately, storage solutions have remained one of the last bastions of proprietary technology, and most of today's unified storage solutions remain largely closed, expensive, and rigid in their capabilities. Most storage appliances are designed as highly-proprietary special-purpose devices, and their design focus often brings about issues that can limit their usefulness. Vendors often old their customers hostage for software fixes and hardware upgrades. Organizations are tied to specific vendor architectures and management platforms that essentially hold their data ransom. Unfortunately, this proprietary approach ultimately puts organizations at risk, making it nearly impossible for them to quickly respond to business needs and capitalize on market opportunities.

A decade ago, servers went through a transition that incorporated industry-standard components and open source software. This transition resulted in increased flexibility for those who used server technology, even as it reduced costs. Now the same shift to open systems is taking place in storage technology, with similar benefits and advantages.

Sun Storage 7000 Unified Storage Systems

Sun created the Sun Storage 7000 Unified Storage Systems to respond to the unprecedented needs for effective, manageable, and scalable data storage. Much more than just another storage appliance, Sun's open-systems approach carefully selects the best general-purpose servers and storage components, combines them with innovative game-changing new technologies, and unifies them with storage software. These systems offer performance and scalability while providing a simple and powerful interface for administrators.

This new and open approach to storage infrastructure lets organizations increase production uptime while reducing the time to troubleshoot storage issues. Resulting systems deliver high performance at lower cost, in less space than competitors, and with simplified management and health of complex storage architecture. Sun's open unified storage systems with integrated enterprise flash technology combine open source software with the OpenSolaris™ Operating System (OS) and Solaris™ ZFS™. The result is scalable infrastructure with intuitive monitoring and management, predictable performance, and flexible services designed to meet the needs of the organization.

In short, Sun Storage 7000 Unified Storage Systems promise to do for storage what open systems did for servers. Ideal for new and fast-growing storage build-outs where organizations want to simplify and speed storage administration, these storage appliances exploit the power of today's multicore, multithreaded processors to increase application performance. Advanced new technology such as solid state devices (SSDs) fundamentally change the appliance memory and bandwidth equation. Together with advanced features such as Solaris DTrace and Solaris ZFS, these appliances provide unprecedented real-time performance and debugging analysis, high computational scalability, strong security, and end-to-end data integrity.

Capabilities brought by Sun Storage 7000 Unified Storage Systems provide:

- ***Quick Installation and Configuration***

Sun Storage 7000 Unified Storage Systems provide up to 10-fold faster installation, requiring only four minutes for a full install. Click-and-drag administration is provided with no training required. An installation wizard takes the guess-work out of tuning the system, and even clustered scenarios are easy to configure and deploy.

- ***Intelligent Analysis and Optimization of the Storage System***

Extremely simple-to-use graphical tools and detailed analytics software are provided. The software provides real-time visibility of CPU, memory, data, data protocol, disk, and network performance. These systems also represent one of the most comprehensive self-healing storage systems. The systems can be rebuilt in minutes, not days, with an instant RAID rebuilder. Performance data can be saved and analyzed for the life of the product.

- ***Easily Scalable Storage Infrastructure***

Sun Storage 7000 Unified Storage Systems are the first appliances to provide Hybrid Storage Pools that optimize the way that storage is spread across memory, solid state devices, and disk storage. Administrators can easily perform non-disruptive adds, moves, or changes to the storage infrastructure. Automated data placement and migration is provided. One software stack across the entire product family means that there is no need for script re-writes when moving from one platform to another. Perhaps best of all, data services can be added easily with with no additional software license to manage or buy.

- ***Rapidly Diagnose, Troubleshoot, and Resolve Issues***

Real-time analytics provided with the appliances let administrators visualize and manage both read and write I/O performance and health. Real-time visibility is provided throughout the system.

- ***Rely on a Comprehensive Portfolio of Services***

Since data storage is such an essential component of IT infrastructure, Sun offers a comprehensive and flexible portfolio of services. Sun's services portfolio helps organizations seamlessly integrate their next-generation unified storage solution into their storage environment, and optimizes its use. With an open-systems focus, Sun also remains active in the community, offering the full spectrum of services to help organizations along the path to open storage.

The World's First Open Storage Appliances

Unlike special-purpose appliance vendors, Sun's unified storage appliances combine tried and tested general-purpose components with open software and innovative new technology. This approach unites the considerable strengths of world-class computational building blocks with enterprise-class OpenSolaris operating system technology, innovative use of enterprise flash technology, and a seamless software environment in the form of Sun Storage 7000 Management software.

Sun Storage 7000 Management Software

Sun Storage 7000 Management software provides the data protocols, data services, and additional management support for the Sun Storage 7000 Unified Storage System. A simple out-of-the-box browser-based GUI gives administrators access to storage management with the rapid familiarity of a point-and-click interface. Much of the appliance's enterprise-class functionality is provided by the OpenSolaris operating system, yielding a rich and proven enterprise-class functionality set that provides the basis for the Sun Storage 7000 Unified Storage System.

Solaris ZFS provides key functionality for the Sun Storage 7000 Unified Storage System, as an enterprise-class, general-purpose file system that provides virtually unlimited file system scalability and increased data integrity to large-scale solutions. Providing up to 21 billion YottaBytes¹ of capacity, this 128-bit, open source file system integrates traditional file system functionality with built-in volume management techniques. By automatically allocating space from pooled storage when needed, Solaris ZFS simplifies storage management and gives organizations the flexibility to optimize data for performance.

1.1 Yottabyte is equal to 1 trillion terabytes, or 10^{24} bytes.

Solaris ZFS provides advanced features and functionality, including:

- Data protection features include a complete suite of snapshot functionality, including essentially unlimited snapshots as well as snapshot replication. Comprehensive file system replication is also provided. For backup, the Network Data Management Protocol (NDMP) is supported. ZFS RAID and RAID-Z is supported.
- Broad data accessibility is provided through multiprotocol support in the OpenSolaris OS, including support for:
 - NFS v3 and v4
 - CIFS
 - iSCSI
 - HTTP
 - WebDav
 - FTP
 - NDMP
 - IP v4 and IP v6
- Data recovery support includes instantaneous backup and restore through the ZFS file system. The NDMP V3 and V4 backup/recovery protocol is supported via both Fibre Channel and SCSI. In addition, a number of backup/restore partners have been certified. Disaster recovery replication is provided, and numerous asynchronous replication modalities are supported.
- High availability is provided through support in the OpenSolaris OS. Solaris Predictive Self Healing software proactively monitors and manages system components to help organizations achieve maximum availability of IT services. Solaris Predictive Self Healing is an innovative capability that automatically diagnoses, isolates, and recovers from many hardware and application faults. The Solaris Fault Manager (FM) and Solaris Service Management Facility (SMF) are the two main components. Clustering of appliances is also supported to enhance high availability.
- Real-time business analytics are provided through DTrace Analytics. This unique functionality gives administrators autonomic real-time system diagnosis capabilities and real-time workload analysis. For example, administrators can see what files are “hot” at any given time, analyze the distribution of reads and writes, and determine active clients displayed by protocol.

Industry-Standard Server and Storage Components

Sun Storage 7000 Unified Storage Systems are constructed from industry-standard server and storage components. This approach takes advantage of Sun’s innovation, and provides a range of capable system configurations. Table 1 lists the capacities and features of Sun Storage 7000 Unified Storage Systems.

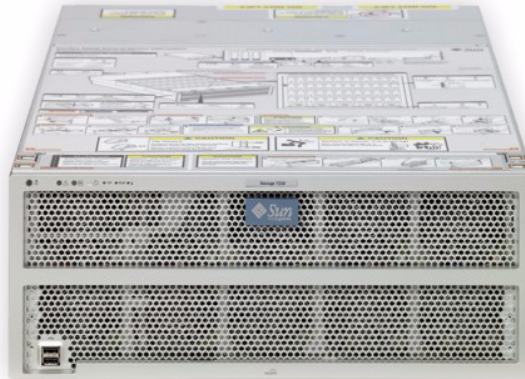
Table 1. Sun Storage 7000 Unified Storage Systems, their capacities and capabilities

Unified Storage Appliance	Capacity	Features
Sun Storage 7110 Unified Storage System	2 TB	• Standalone appliance
Sun Storage 7210 Unified Storage System	Up to 48 TB	• Standalone appliance
Sun Storage 7410 Unified Storage System	Up to 575 TB	• Standalone appliance or clustered for high availability • Enterprise SSDs for enhanced performance • Support for multiple Sun Storage J4400 expansion arrays

Figure 1 illustrates the Sun Storage 7000 Unified Storage System family.



Sun Storage 7110 Unified Storage System



Sun Storage 7210 Unified Storage System



Sun Storage 7410 Unified Storage System

Figure 1. Sun Storage 7000 Unified Storage System family

Sun Storage 7110 and 7210 Unified Storage Systems operate as standalone storage appliances. The Sun Storage 7410 Unified Storage System can be augmented with additional Sun Storage J4400 storage arrays, and enterprise SSDs. Clustered configurations of the Sun Storage 7410 Unified Storage System are also supported for high availability. No matter what Sun Storage 7000 Unified Storage System is selected, services and warranty are provided to cover both the hardware and software.

Chapter 2

Key Storage Technology Innovations

With flat to declining storage budgets, many organizations are searching for imaginative alternatives to meet their data-growth challenges. Sun recognizes that the computing, storage, and networking are converging, and is keenly focused on bringing about technology innovations that make the most of open systems, innovative new technology, and industry-standard building blocks.

Coupled with open systems, open storage, and open networks, Sun Storage 7000 Unified Storage Systems take advantage of OpenSolaris innovations as well as recent advances in the state of enterprise solid state storage devices (SSDs). Solaris ZFS and its Hybrid Storage Pool technology is an ideal match for enterprise SSD technology, allowing the pooling of memory, SSDs, and hard disk drives. DTrace Analytics provides in-depth monitoring of the complete storage system.

Solaris™ ZFS™

Sun Storage 7000 Unified Storage Systems rely heavily on Solaris ZFS for key functionality such as Hybrid Storage Pools. By automatically allocating space from pooled storage when needed, Solaris ZFS simplifies storage management and gives organizations the flexibility to optimize data for performance. Key capabilities of Solaris ZFS related to the hybrid storage pool include:

- *Virtual storage pools* — Unlike traditional file systems that require a separate volume manager, Solaris ZFS introduces the integration of volume management functions. Breaking free of the typical one-to-one mapping between the file system and its associated volumes, Solaris ZFS introduces the storage pool model. Solaris ZFS decouples the file system from physical storage in the same way that virtual memory abstracts the address space from physical memory, allowing for more efficient use of storage devices. Space is shared dynamically between multiple file systems from a single storage pool, and is parceled out of the pool as file systems request it. Physical storage can be added to storage pools dynamically, without interrupting services. When capacity is no longer required by one file system in the pool, it becomes available to other file systems.
- *Data integrity* — Solaris ZFS uses several techniques to keep on-disk data self consistent and eliminate silent data corruption, such as copy-on-write and end-to-end checksumming. Data is written to a new block on the media before changing the pointers to the data and committing the write. Because the file system is always consistent, time-consuming recovery procedures like `fsck(1)` are not required if the system is shut down in an unclean manner. In addition, data is read and checked constantly to help ensure correctness, and any errors detected in a mirrored pool are automatically repaired to protect against costly and time-consuming data loss and

previously undetectable silent data corruption. Corrections are made possible by a RAID-Z implementation that uses parity, striping, and atomic operations to aid the reconstruction of corrupted data.

- *High performance* — Solaris ZFS simplifies the code paths from the application to the hardware, delivering sustained throughput at near platter speeds. Block allocation algorithms accelerate write operations, and consolidate many small random writes into a single, more efficient sequential operation. Indeed, an I/O scheduler bundles disk I/O to optimize arm movement and sector allocation to speed throughput. In addition, an intelligent prefetch performs read ahead for sequential data streaming, and can adapt its read behavior on the fly for more complex access patterns. Furthermore, data is striped automatically across all available storage devices to balance I/O and maximize throughput. Solaris ZFS immediately begins to allocate blocks from devices as soon as they are added to the storage pool, increasing effective bandwidth as each device is added to the system.
- *Simplified administration* — Solaris ZFS automates many administrative tasks to speed performance and eliminate common errors. Creating file systems is fast and easy. There is no need to configure, or reconfigure, underlying storage devices or volumes — these tasks are handled automatically when devices are added to a storage pool. In addition, administrators can guarantee a minimum capacity for file systems, or set quotas to limit maximum sizes.

Enterprise Solid State Storage Devices

Nearly everyone is familiar with some sort of commercially available Flash device, from memory cards used in MP3 players, cell phones, and digital cameras to store music, photographs, and other digital information, to removable USB drives used to backup and transport data from one machine to another. Technological advancements are moving NAND Flash technology past simple commodity use and making it a reasonable storage alternative for the enterprise. Robust data integrity, reliability, availability and serviceability features, combined with breakthrough performance and power characteristics, have made it possible to create a new class of storage device.

Enterprise solid state devices based on Flash technology consist of three main components: NAND Flash, DRAM, and a controller (Figure 2).

- **NAND Flash**

NAND Flash is used for primary back-end storage, and requires blocks to be erased prior to writing data. While NAND Flash provides fast read access times, it takes 1.5 milliseconds to erase a block. Maintaining a range of spare blocks that are available for use helps to alleviate erase time bottlenecks.

Enterprise SSDs are available in both *read-biased* and *write-biased* versions to accelerate different operations within Sun Storage 7000 Unified Storage System

- **DRAM**

DRAM provides a local buffer to accelerate Flash write performance and maintain active data structures. A supercapacitor is used to protect data and permit it to be flushed to the media in the event of power loss.

- **Controller**

A controller manages the back-end storage and buffer cache, and provides a communication interface to systems. To extend the life of the device, the controller works to minimize writes to the same location to decrease wear, and tracks and maps bad blocks so they cannot be reused. While mapping bad blocks out of the available address space can impact performance over time, doing so results in gradual device failure rather than sudden failure. In addition, information is load balanced and interleaved to speed performance, and ECC is supported to provide data integrity.

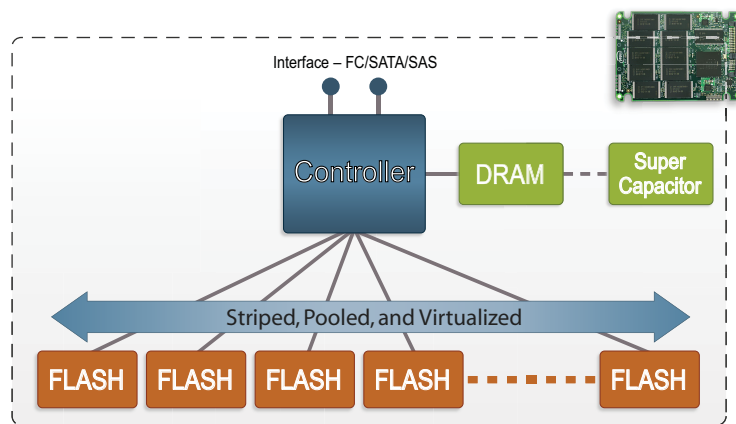


Figure 2. Enterprise SSD high-level architecture

Several advancements in Flash technology characteristics are making it possible to utilize SSDs in the enterprise datacenter.

- **Performance**

Flash technology completes operations in microseconds, placing it between hard disk drives (milliseconds) and DRAM (nanoseconds) for access time. Because Flash technology contains no moving parts, it avoids the seek times and rotational latencies associated with traditional hard disk drives. As a result, data transfer throughput to and from solid state storage media is faster than electromechanical disk drives can provide — with enterprise SSDs providing tens of thousands of IOPS compared to hundreds of IOPS for hard disk drives.

- **Low Power Consumption**

Hard disk drives draw significant amounts of power to operate the motor and spin the media. In contrast, the use of efficient Flash integrated circuits and a lack of motors and other mechanical parts result in enterprise SSDs consuming a fraction

of the power of conventional hard disk drives. In fact, enterprise SSDs use only 5% of the power used by hard disk drives when idle, and as little as 15% when performing operations. As a result, enterprise SSDs produce less heat in the system chassis.

- **Cost**

While Flash devices are more expensive per gigabyte than a comparable disk drive, Flash memory costs are dropping significantly year over year. In addition, as electricity costs rise and Flash memory costs decrease, the relative cost per available gigabyte and cost per IOPS of Flash memory improves. For example, hard disk drives cost approximately \$1.25/IOPS, compared to only \$0.02/IOPS for enterprise SSDs. Since hard disk drives must be powered on to be available, the low power consumption of enterprise SSDs makes them a smart choice for datacenters looking to reduce operating costs. While a gigabyte of mechanical disk costs less than a gigabyte of Flash memory, the fact that Flash memory outperforms hard disk storage by at least an order of magnitude in reading and writing data makes the cost per gigabyte of Flash devices exceptionally low.

- **Reliability**

While enterprise SSDs provide similar functionality to traditional hard drives, they offer improved reliability features. Both hard disk drives and enterprise SSDs support bad block management, wear leveling, and error correction codes (ECC) to foster data integrity. However, unlike hard drives that use a motor to spin magnetic media and a read/write head that must move to perform operations, enterprise SSDs contain no moving parts — data is stored on integrated circuits that can withstand significant shock and vibration. In fact, enterprise SSDs operate in a wider thermal operating range and wider operational vibration range than hard disk drives to deliver significantly higher Mean Time Between Failure (MTBF) (2.0 million hours versus 1.2 million hours).

SSDs are now available in a form factor similar to hard disk drives, allowing them to be deployed in place of actual hard disk drives. SSDs are also available in versions that are optimized for reading or writing. In Sun Storage 7410 Unified Storage systems, these different types of SSDs are used for different purposes:

- **Write-biased SSDs**

Write-biased SSDs are used in place of NVRAM to host the ZFS Intent Log (ZIL). Writes are buffered by DRAM backed by supercapacitors. In clustered configurations, write-biased SSDs are placed in Sun Storage J4400 expansion arrays instead of a slot in head node. With this approach, multiple head-nodes in clustered environments have access to the write cache. As of this writing, up to 12 18GB write-based SSDs are supported.

- **Read-biased SSDs**

Read-based SSDs are placed in the server node in order for cache hits to have the shortest possible return route to the network adapter. Read-based SSDs are used to extend the ZFS cache (L2ARC) for reads and writes. As of this writing, up to 600 GB of read-biased SSDs are supported.

Solaris ZFS and Hybrid Storage Pools

Solaris ZFS provides a seamless and easy way to administer hybrid storage pools, taking advantage of the performance of enterprise-class SSDs and inexpensive hard disk drive capacity. Unlike traditional volume managers that simply use SSDs in a RAID stripe, Solaris ZFS integrates the volume manager with the file system and can use enterprise SSDs more effectively. For example, Solaris ZFS can use SSDs intelligently as a cache for both application and file system metadata, placing latency-critical data structures appropriately on Flash media and using algorithms to optimize data placement. In addition, Solaris ZFS provides acceleration of both read and write operations, and lets administrators configure the system to match workload demands (Figure 3). These concepts are explored in the sections that follow.

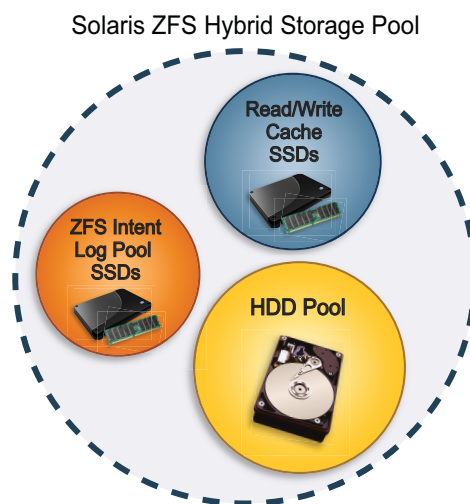


Figure 3. Solaris ZFS automates storage management and helps balance system performance with the ZFS intent log, a read cache, and a high-capacity storage pool

ZFS Hybrid Storage Pool Architecture

Both read-biased and write-biased SSDs are used by ZFS to accelerate the performance of Hybrid Storage pools in Sun Storage 7000 Unified Storage Systems. Figure 4 illustrates how various key components of the ZFS architecture are deployed across a Hybrid Storage Pool.

- The ZFS Adaptive Replacement Cache (ARC) is the main ZFS memory cache in DRAM.
- The Level Two Adaptive Replacement Cache (L2ARC), extends the ARC into read-biased SSDs to provide a large read cache to accelerate reads.

- The ZFS Intent Log (ZIL) is transactional, and uses write-based SSDs to provide a large cache to accelerate writes
- The disk storage pool consists of conventional disk drives. Note that especially fast disk drives are no longer strictly required given the interposition of SSDs in the Hybrid Storage Pool.

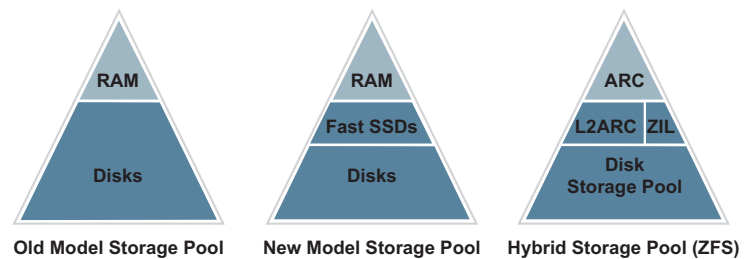


Figure 4. Solaris ZFS Hybrid Storage Pool Architecture distributes key functionality into RAM, SSDs, or disk drives as appropriate.

Read-Biased SSDs for Reduced Read Latency

Systems use memory to cache frequently-accessed data for rapid access and improved performance. Once data is stored in the cache, future requests can be satisfied quickly by accessing the cached copy rather than fetching it from disk. Policies determine the data that is held in the cache in an attempt to anticipate future needs. However, large working sets that cannot fit into memory can cause the cache to be ineffective.

Flash storage can be used to enhance caching operations in systems. Solaris ZFS combines main memory and enterprise SSDs into a large read cache and uses an Adaptive Replacement Cache (ARC) for its cache replacement algorithm. The ARC manages and balances the cache content using most frequently used (MFU) and most recently used (MRU) algorithms for storing data to, and retrieving data from, memory. A second-level ARC (L2ARC) with smart caching and pre-fetching techniques lets Solaris ZFS use enterprise SSDs as a second-level cache to further speed read performance. Defective Flash blocks are treated as a cache miss rather than data loss, with information retrieved from hard disk to satisfy the request. The checksums built into Solaris ZFS are used to catch cache inconsistencies. Using Solaris ZFS and an L2ARC stored on Flash devices helps:

- **Eliminate Disk Latency**

Both the ARC and L2ARC are used to satisfy read requests from clients, and aim to avoid blocking a read request due to disk latency. Read operations can be serviced by the combined caches rather than disk drives. As a result, applications block for no more than the duration of Flash latency (< 100 us) rather than the latency of disk drives (up to 4 ms).

- ***Speed Access to Working Sets***

Flash devices offer a faster way to access working sets that do not fit into available memory. While Flash devices are more expensive than fast hard disk drives per unit of storage, caching a very large working set on Flash devices costs less than storing all the data on fast disks when the performance advantages of Flash technology are taken into account.

- ***Enhance Cache Performance***

The L2ARC uses an evict-ahead policy. Cache entries are aggregated and predictively pushed out to Flash devices in order to distribute overhead across large write operations and eliminate additional latency that could arise when an entry is evicted from the cache. A ring buffer is used to manage the L2ARC replacement policy. When the end of the cache is reached, entries are stored at the beginning of the cache to avoid potential fragmentation. While it is possible for entries to be overwritten in the L2ARC prematurely, the most frequently accessed data still resides in DRAM-based cache.

- ***Speed System Readiness by Warming Caches***

The L2ARC stores a directory of data blocks written to the L2ARC. This practice helps to identify cache contents after a power or system failure and warm the cache. Instantly warming the cache reduces the time needed to restore production systems after planned or unplanned outages or other system resets.

- ***Reduce the Volatility of Cache Content***

Since the L2ARC writes to flash devices slowly, and data on the system can be modified very quickly, it is possible for the contents of the L2ARC to be different than the data stored on disk. During normal operation, dirty and stale entries are marked and ignored. After a system reset, stale data can be read from cache devices. However, metadata kept on the device, and the checksums built into Solaris ZFS, are used to identify this condition and seamlessly recover by reading the correct data from disk.

Write-Biased SSDs for Reduced Write Latency

Solaris ZFS uses a log to record modifications to the file system. The Solaris ZFS Intent Log (ZIL) allows applications that demand synchronous writes to a permanent storage medium to benefit from apparent latency reductions, and get work done while data is written asynchronously in the background. The ZIL can store small transactions to the file system in a dedicated enterprise SSD pool before committing the transaction to disk. The ZIL also stores enough information to replay the transaction, if needed. These records are freed after the data is committed to disk.

The ZIL handles small and large writes differently.

- Small writes are included in the log record.
- Large writes are synchronized to disk, and the ZIL maintains a pointer to the synchronized data in the log record. As a result, the size of the ZIL tends to be small and is dictated by the number of IOPS from clients.

Several techniques are used to speed write throughput.

- Solaris ZFS manages the storage pool by aggregating high-bandwidth devices and low-latency devices separately. It dynamically determines whether a low-latency or high-bandwidth device should be used, depending on the amount of accumulated data in a transaction.
- Writes are acknowledged once the data is written to the ZIL. Multiple small transactions are aggregated, letting the system perform fewer commits to the hard disk drives in the storage pool and use fewer and larger I/O transactions to speed I/O throughput. The file system writes uniformly to each byte in the intent log SSD to help alleviate flash wear out.
- Placing the ZIL on a low-latency enterprise SSD can help improve server throughput. For example, internal testing revealed ZIL latency in the 80 us to 100 us range. In this configuration, Solaris ZFS wrote the ZIL to the enterprise SSD in 8 KB chunks, with each write completing in 80 us — far faster than the milliseconds needed to access a hard disk drive.

Detailed DTrace Analytics

DTrace is a dynamic tracing framework that helps organizations simplify the process of identifying the source of intermittent and sustained application performance problems. With DTrace, administrators and application developers can instrument a live operating system kernel and running applications without rebooting the kernel or recompiling — or even restarting — applications. Instrumentation can be activated as desired, leaving no overhead when tracing is turned off.

In Sun Storage 7000 Unified Storage Systems, DTrace Analytics provides real-time observability for:

- NFS V3 and V4
- CIFS
- iSCSI
- ZFS
- CPU
- Memory utilization
- Networking

The in-depth analytics provided with DTrace probes are key to helping organizations fine-tune their unified storage appliances. For instance, these analytics can help administrators understand and optimize their workloads in real time, helping them to:

- Understand the benefits of write-biased and read-biased SSDs on their specific storage workloads
- Understand when CPU, memory, and networking are providing bottlenecks
- Understand the read/write/metadata mix of their particular workloads

More information on real-time dashboard and analytics features is provided in *Chapter 4*.

Chapter 3

Sun Storage 7000 Unified Storage Systems

Sun Storage 7000 Unified Storage Systems are based on general-purpose platforms, but deliver the advantages of unified storage appliances. The sections that follow describe individual Sun Storage 7000 Unified Storage Systems and their capabilities.

Sun Storage 7110 Unified Storage System

Based on a two rack unit (2U) industry-standard server platform, the Sun Storage 7110 Unified Storage System provides up to 2 TB of storage capacity. This standalone NAS appliance provides processors, memory, and storage combined into a convenient unified storage platform. Features of the Sun Storage 7110 Unified Storage System include:

- Standard storage pools (no SSDs)
 - 8 GB of memory
 - Sixteen 146 GB SAS 10K RPM disk drives
- Network connectivity
 - Standard 4 x 1 Gb Ethernet ports
 - Optional dual-port 10 Gb Ethernet (copper or optical)
 - Optional quad-port 1 GB Ethernet
- Tape backup connectivity
 - Optional dual-port Fibre Channel HBA
 - Optional dual-port SCSI HBA

As shown in Figure 5, all 16 hot-plug disk drives of the Sun Storage 7110 Unified Storage System are accessed from the front of the chassis.



Figure 5. The Sun Storage 7110 Unified Storage System supports capacities up to 2 TB.

Sun Storage 7210 Unified Storage System

Supporting up to 48 TB, the Sun Storage 7210 Unified Storage System offers considerable standalone storage capacity in a 4U rackmount configuration. Write-biased SSDs are also supported for reduced write latency. Features of the Sun Storage 7210 Unified Storage System include:

- Hybrid Storage Pools
 - 32 GB or 64 GB of RAM
 - Write-biased SSDs, up to 2 x 18 GB
 - Up to 48 250 GB, 500 GB, or 1 TB SATA disk drives
- Network connectivity
 - Standard 4 x 1 Gb Ethernet ports
 - Optional dual-port 10 Gb Ethernet (copper or optical port 10 Gb Ethernet)
 - Optional quad-port 1 Gb Ethernet
- Tape backup and connectivity
 - Optional dual-port Fibre Channel HBA
 - Optional dual-port SCSI HBA

As shown in Figure 6, the Sun Storage 7210 Unified Storage System provides space for up to 48 top-loading SATA disk drives.



Figure 6. The Sun Storage 7210 Unified Storage System provides up to 48 TB of capacity

Sun Storage 7410 Unified Storage System

The Sun Storage 7410 Unified Storage System represents an expandible unified storage appliance that can be clustered and augmented with additional JBOD (just a bunch of disks) storage arrays. The basic server building block for the appliance is a 2U rackmount server with support for up to eight internal disk drives and considerable memory capacity.

Features of the Sun Storage 7410 Unified Storage System include:

- Hybrid Storage Pools
 - 16 GB, 64 GB, or 128 GB of RAM
 - Up to twelve 18 GB write-biased SSDs
 - Up to six 100 GB read-biased SSDs
 - Up to 288 1 TB SATA disk drives
 - Up to 12 Sun Storage J4400 arrays with 24 hard disk drives each
- Network connectivity
 - Standard 4 x 1 Gb Ethernet ports
 - Optional dual-port 10 Gb Ethernet (copper or optical)
 - Optional quad-port 1 Gb Ethernet
- Tape backup and connectivity
 - Optional dual-port Fibre Channel HBA
 - Optional dual-port SCSI HBA

As shown in Figure 7, an additional clustered server can be added for high availability. Up to 12 additional Sun Storage J4400 arrays can be added for increased capacity.



Figure 7. Sun Storage 7410 Unified Storage Systems can be configured in clustered configurations, and capacity can be expanded with up to twelve 24-disk Sun Storage J4400 arrays.

Expansion Arrays

The Sun Storage J4400 array offers enterprise-class JBOD capabilities. With dual SAS controllers for robust connectivity options, each array delivers up to 24 3.5-inch SATA devices in only four rack units. Independently hot-swappable drives are front-accessible for easy replacement.

In the Sun Storage 7410 Unified Storage System, these expansion arrays can be configured either full or half-full, and each can be configured with up to four write-biased SSDs. A variety of supported configuration options are available for Sun Storage J4400 arrays configured into the Sun Storage 7410 Unified Storage System. Table 2 lists the supported configurations, including both data disks and write-biased SSDs.

Recommendations for specific workloads are beyond the scope of this document. However, the detailed DTrace Analytics provided by Sun Storage 7000 Unified Storage Systems can prove very helpful in making specific configuration decisions based on actual storage workloads.

Table 2. Supported configurations for Sun Storage J4400 expansion arrays as a part of Sun Storage 7000 Unified Storage Systems.

JBOD Configuration Designation	Data Disks	Raw Data Capacity	Write-Biased SSDs	SSD Capacity
J4400 Array Half 0	12	12 TB	0	0 GB
J4400 Array Half 1	11	11 TB	1	18 GB
J4400 Array Half 2	10	10 TB	2	36 GB
J4400 Array Full 0	24	24 TB	0	0 GB
J4400 Array Full 1	23	23 TB	1	18 GB
J4400 Array Full 2	22	22 TB	2	36 GB
J4400 Array Full 4	20	20 TB	4	72 GB

Figure 8 illustrates two Sun Storage 7410 Unified Storage System configurations. The left-most configuration consists of a head node and a single expansion array. The configuration on the right employs three expansion arrays connected in a single daisy chain configuration.

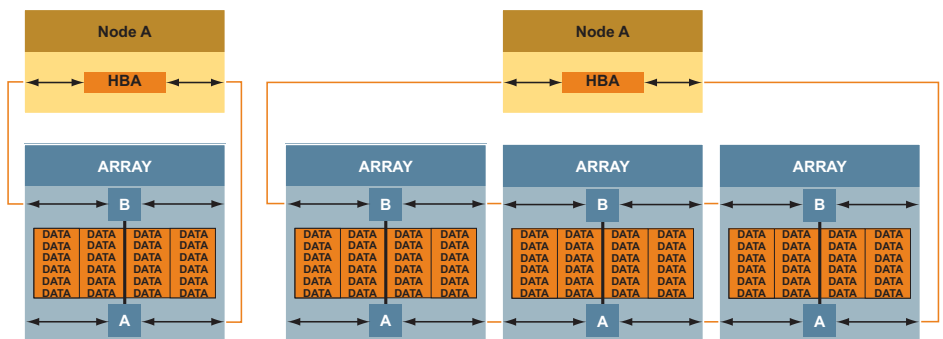


Figure 8. Example Sun Storage 7410 Unified Storage System with single array (left) and single array daisy chain (right)

To increase resiliency to HBA failures, Sun Storage 7410 Unified Storage Systems can also be configured with multiple array daisy chains as shown in Figure 9. In the event of a single HBA failure, or a cabling disconnect, the appliance can continue to function in this configuration.

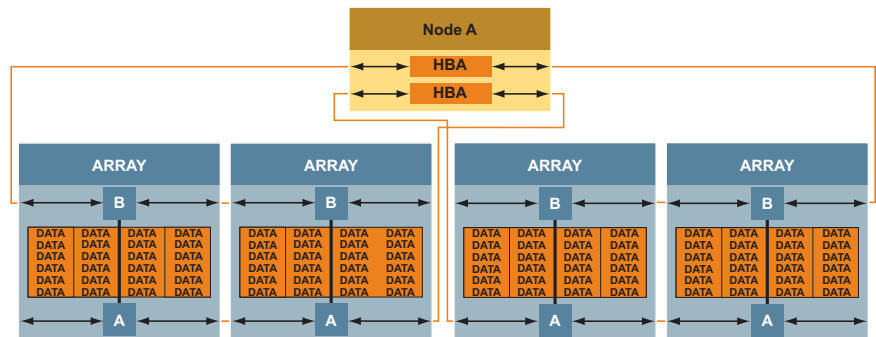


Figure 9. The Sun Storage 7410 Unified Storage System with multiple array daisy chains is resistant to HBA failure

Clustered Configurations for High Availability

The Sun Storage 7410 Unified Storage System can be configured in a clustered configuration to enhance availability. With active/passive clustered configurations, a second server node is provided and configured as a hot spare — providing for fail-over and protecting against the loss of a server node. Active-active configurations are also supported. In addition, if multiple expansion arrays are purchased initially, ZFS can be configured to perform RAID-Z across the JBOD expansion arrays themselves for resilience against the failure of an entire expansion array. Figure 10 illustrates clustered configurations with both single and multiple expansion arrays.

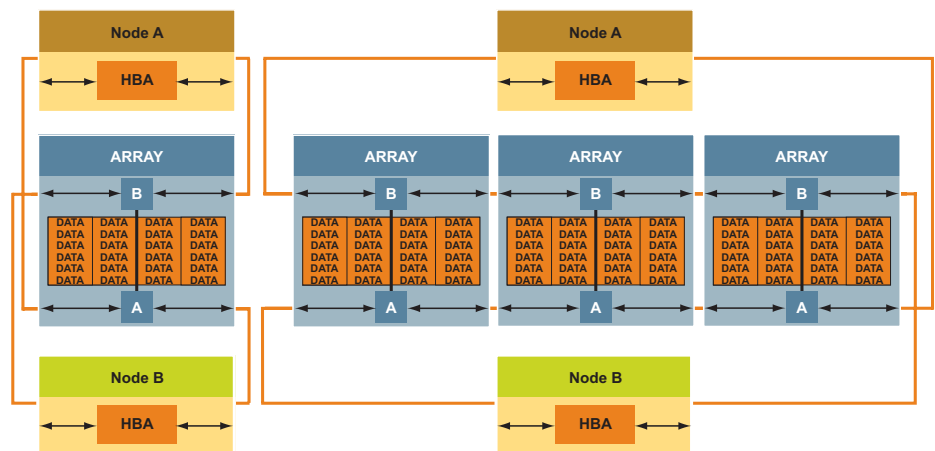


Figure 10. Clustered Sun Storage 7410 Unified Storage System configurations with single (left) and multiple (right) expansion arrays

Figure 11 shows a clustered configuration with multiple expansion arrays and both read-biased and write-biased SSDs installed. It is important to note that in a clustered configuration, write-biased SSDs are placed in the expansion arrays instead of a slot in

head node. With this approach, multiple head-nodes in clustered environments have access to the write cache. In contrast, read-biased SSDs are placed in the server nodes in order for cache hits to have the shortest possible return route to the network adapter.

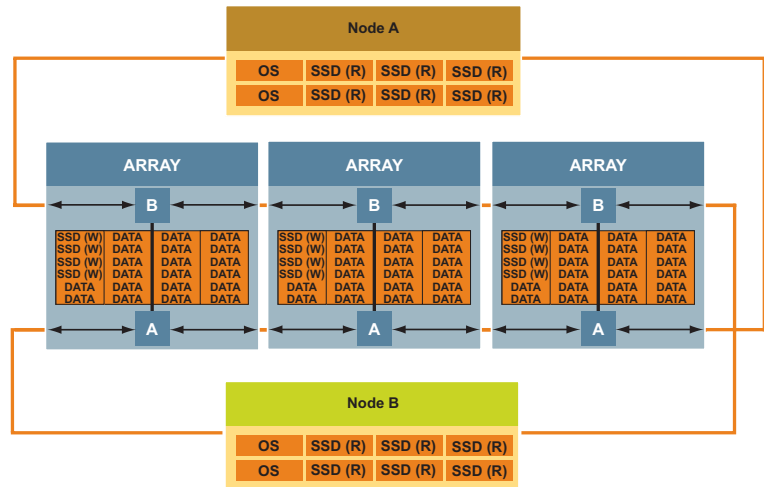


Figure 11. A clustered Sun Storage 7000 Unified Storage System Clustered configuration is shown with read-biased and write-biased SSDs installed.

Chapter 4

Sun Storage 7000 Management Software

Most NAS and unified storage solutions come with proprietary software offerings that impose arbitrary operational limitations. Software is often a primary revenue source for appliance vendors, and complicated licensing structures and fees can add considerably to the cost of storage solutions. In contrast, Sun Storage 7000 Storage Management software provides a full complement of storage software that allows organizations the flexibility to use the appliance as their needs dictate. Unlike competitor's products, Sun Open Storage Management software is included with the system at no additional cost.

Real-Time Dashboard and Advanced Analytics

In keeping with the appliance concept for Sun Storage 7000 Unified Storage Systems, a real-time dashboard (Figure 12) acts as an administrative home page for the system. Even though these unified storage systems are based on general-purpose Sun servers, all administration takes place through the real-time dashboard. This Web-based graphical user interface (GUI) provides continuous monitoring of key performance metrics and provides a real-time feed of relevant alerts. Administrators can use the dashboard for at-a-glance usage statistics, and as an entry point to more advanced DTrace Analytics for the appliance.

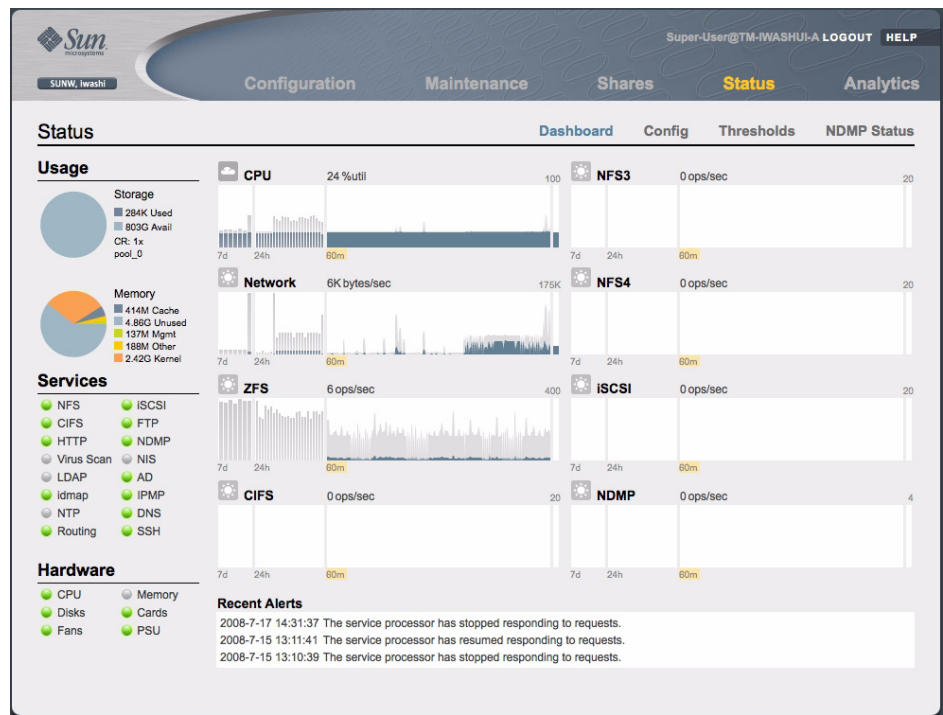


Figure 12. The real-time dashboard provides continuous monitoring of key performance metrics

An easy-to-learn command-line interface (CLI) is also provided, and is accessed over SSH for security. Input commands execute locally on the appliance to change its configuration. All of the features found in the Web-based GUI are present in the CLI, except for the graphical DTrace Analytics and dashboard elements. CLI scripting is also available, and scripts can be written and sent over SSH to the appliance. CLI scripting can be used to integrate with end-user scripts. For instance, a script might create a home directory for each user in an LDAP directory. Underlying XML-RPC calls are also exposed to developers.

More than just an administrative dashboard, the GUI provides extensive advanced real-time DTrace Analytics not found in competing products. Implemented on top of the DTrace facility in OpenSolaris, organizations can use these analytics to drill down and collect real-time statistics on all aspects of the appliances' operation. Queries can be saved and executed in the background. Results can be saved on the appliance, or exported. Table 13 illustrates an example screenshot showing this functionality.

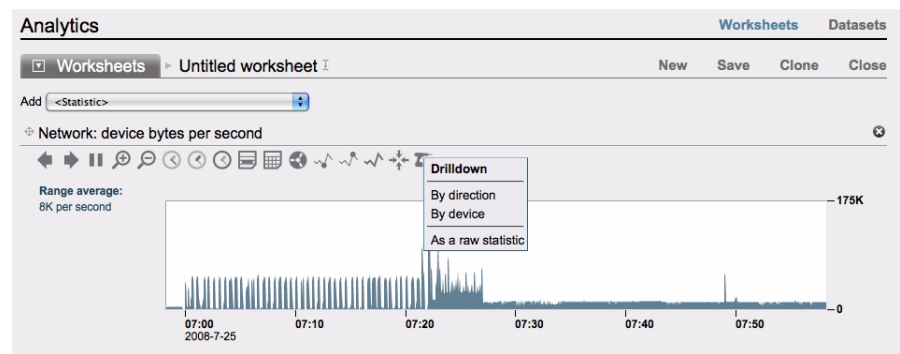


Figure 13. Administrators can use the dashboard GUI to drill down into appliance statistics

DTrace Analytics are provided for 29 system statics in nine categories as shown in Table 3.

Table 3. Categories for DTrace Analytics and drill-down subcategories

Category	Statistic	Drill-Down Subcategories
CIFS	Operations	Op, Client, Filename, Share, Project, Latency, Size, Offset, Raw
CPU	Utilization Percent	CPU Mode, CPU ID, Application, Process ID, User ID, Raw
	Kernel Spins	CPU Mode, CPU ID, Application, Process ID, User ID, Raw
Disk	iSCSI Operations	Op, Client, Raw
	I/O Operations	Type, Disk, Size, Latency, Offset, Raw
	Utilization Percent	Disk, Raw
	ZFS I/O Bytes	Op, Pool, Raw
	ZFS I/O Operations	Op, Pool, Raw
File system	DNLC Entries	Raw
	DNLC Accesses	Hit/Miss, Raw
HTTP/ WebDAV	Requests	Op, Result, Client, Filename, User, User Agent, Size, Latency, Raw
Memory	Dynamic Memory Usage	Application, Raw
	Kernel Memory	Kmem Cache, Raw
	Kernel Memory in Use	Kmem Cache, Raw
	Kernel Memory Lost to Fragmentation	Kmem Cache, Raw
NDMP	Disk Bytes Transferred	Op, Raw
	File system operations	Op, Raw
	Jobs	Op, Raw
	Tape Bytes Transferred	Op, Raw
Network	Device Bytes	Direction, Device, Raw
	Interface Bytes	Direction, Interface, Raw
	IP Bytes	Hostname, Protocol, Direction, Raw
	IP Packets	Hostname, Protocol, Direction, Raw
	TCP Bytes	Hostname, Protocol, Direction, Raw
	TCP Packets	Hostname, Protocol, Direction, Raw
ZFS	ARC Adaptive Parameter	Raw
	ARC Accesses	Hit/Miss, Raw
	ARC Size	Raw
	ARC Target Size	Raw

Data Protocols

In order for a unified storage appliance to have maximum utility, it must support as many storage protocols as possible. Sun Storage 7000 Unified Storage Systems work with a large range of data protocols, allowing organizations to use those protocols that make sense for their IT needs.

- *NFS file sharing* — Support for NFS v3 and v4 is provided, along with support for UNIX access control lists (ACLs).
- *CIFS and SMB file sharing* — Shares can be exported to Microsoft Windows and Samba clients using the CIFS protocol.
- *iSCSI shares* — The appliances benefit from OpenSolaris iSCSI target support, and its integration into ZFS. Support is provided for access control lists, CHAP authentication, iSNS clients, multiple clients per session, error recovery levels one and two, data and header digests, and persistent group reservations.
- *HTTP file sharing* — Support is provided for HTTP file access on a per-share basis using the provided Apache Web server and the WebDAV protocol.
- *FTP access to file shares* — FTP access to data is supported, and FTP user access is controlled through the associated directory service.
- *NDMP appliance backups* — Support for the Network Data Management Protocol (NDMP) is provided. Both NDMP v3 and NDMP v4 are supported and NDMP activities and statistics are available in the administrative GUI. Sun StorageTek™ tape drives and virtual tape libraries (VTLs) are also supported. Figure 14 illustrates the GUI for administering NDMP backups.

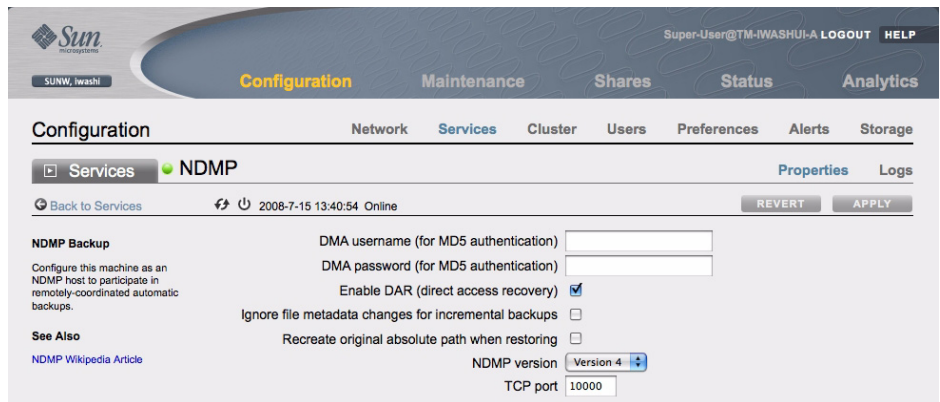


Figure 14. NDMP appliance backups can be administered from within the unified appliance GUI

Data Services

Part of being able to serve large storage needs is being able to manage storage data to meet the needs of IT clients. Beyond acting as a unified storage appliance, Sun Storage 7000 Storage Systems offer a rich set of data services that help extend, manage, and protect valuable storage data.

ZFS NAS Offerings

With its 128-bit foundation, ZFS offers effectively unlimited capacity. Flexibility is provided through thin provisioning that allows the use of dynamic reservations and quotas for projects and shares. With these tools, administrators can both scale and manage storage allocated to various projects that share the appliance. As with all ZFS deployments, end-to-end checksumming of all data and metadata is provided, eliminating the unpredictable effects of silent data corruption. Multiple data protection schemes are provided, allowing administrators to make intentional trade-offs as they tune specific shares for capacity, availability and performance (Table 15).

Configure Storage ◀ Step 2 of 2 ▶				
Configure available storage using a replication profile.				
TYPE ^	CAPACITY	AVAILABILITY	PERFORMANCE	SIZE
Double Parity RAID	○	●	●	1.05T
Mirrored	○	●	●	808G
Single Parity RAID	●	○	●	1.31T
Striped	●	○	●	1.84T

Figure 15. ZFS configuration options allow administrators to tune individual shares for capacity, availability, and performance.

Appliance-to-Appliance Remote Replication

Sun Storage 7000 Unified Storage Systems provide a facility for secure, asynchronous replication of the contents of the system to another appliance. The remote replication facility is based on the ZFS send/receive functionality provided in the base OpenSolaris OS. Data is transmitted securely using a private SSL connection established between two Sun Storage 7000 Unified Storage Systems. Remote replication can be initiated on-demand, executed continuously between two appliances, or scheduled to occur on some regular basis. The progress of active remote replication transfers is visible through the administrative dashboard interface.

Administrators can also specify a particular network connection to use for remote replication. Replications are transactionally consistent, as only differences are sent. Options are available for multiple replication schemes. Administrators can replicate

one-to-one, many-to-one, or one-to-many. Fail-over (receiver) and push-over (sender) are supported for disaster recovery scenarios. Figure 16 illustrates the Web-based GUI for configuring remote replication targets.

Figure 16. Remote replication provides considerable flexibility and security.

Off-Appliance Virus Scanning

Sun Storage 7000 Unified Storage Systems include an Internet Content Adaption Protocol (ICAP) virus scanning utility. The utility can connect to one or more virus scanning engines on other hosts, such as those provided by Symantec and McAfee. The virus scanner can periodically scan, repair, and quarantine files on the appliance shares. Quarantined files are marked to be inaccessible by clients, and lots of virus scans are kept on the appliance.

Snapshots

ZFS provides the Sun Storage 7000 Unified Storage System with an unlimited number of snapshots and writable snapshots. Snapshots can be run manually, or scheduled, and snapshots can also be exported as a new share. Figure 17 illustrates the GUI for administering snapshots.

NAME	CREATION	UNIQUE	TOTAL	CLONES
test	2008-8-14 01:21:16	0	18K	-
Accounting_Minute_5	2008-8-14 01:19	0	18K	-
Account2	2008-8-14 01:19:40	0	18K	-

Figure 17. The dashboard GUI allows for the easy administration of snapshots.

Compression

Sun Open Storage Management software provides four levels of share and iSCSI compression, including:

- LZJB (Fastest)
- GZIP-2 (Fast)
- GZIP (Default)
- GZIP-9 (Best Compression)

Out-of-the-Box Setup and Services Management

Despite being built from general-purpose components, Sun Storage 7000 Unified Storage Systems are shipped with software pre-installed from the factory. Each system is configured for easy out-of-the-box setup. Steps for setting installing each appliance include:

- Installing the hardware and configuring the primary network interface through the serial port
- Starting the browser interface. A secure browser interface is included based on AJAX. Supported browsers include Firefox, Safari, Internet Explorer 6 and 7, and Opera.
- Configuring networking (including name services and the system clock). Supported name services include DNS and NIS. Time and date on the appliance can be configured to automatically update through NTP, or they can be managed manually.
- Optionally configuring one or more directory services. Supported directory services include LDAP, NIS, and Active Directory. LDAP and NIS can control share access and appliance management. Active Directory can only control access to shares, administrative accounts have to be created locally on the appliance for appliance management.
- Configuring storage pools
- Configuring NAS features.

Clustering, RAS, and Management

Reliability, availability, and serviceability (RAS) features are key for storage appliances, helping to ensure that essential data remains available and retains its integrity. Sun Open Storage Management software provides extensive RAS features.

Clustering

Sun Storage 7410 Unified Storage Systems provide clustering support, offering the capability for head-node fail-over for clustered appliances. The clustering software is designed to be extremely simple and fast, supporting fail-over between two systems. The clustering software executes a heartbeat over one or more redundant, dedicated

heartbeat links, and triggers fail-over either on-demand, or when heartbeat failure is detected. Sun Open Storage Management software provides the following clustering functionality.

- *Guided Cabling* — To create a cluster, administrators need only cable a single heartbeat link, provide power to the second head node, and then issue an administrative command from the Web dashboard for the first head node to add the second node. The heartbeat link itself is used to transmit relevant configuration data to the second node, while the Web dashboard provides a guided experience.
- *Synchronized administrative configuration* — Sun Open Storage Management software provides for synchronized appliance configurations. When Systems Management Facility (SMF) changes occur on one side of the cluster, these changes are replicated to the other side of the cluster. If a cluster head-node reboots or is down for some extended period of time, it will re-synchronize its SMF configuration.
- *Fail-over for network resources* — With clustering support, shared network interfaces are pre-configured on both sides of the cluster, with only those of the active head-node configured as “up”. When fail-over occurs, the clustering software automatically “brings up” the interfaces on the new active node, and issues network requests to claim the address and update other entities on the network with its physical Ethernet MAC address.
- *Fail-over for storage resources* — To support the storage configurations described for the Sun Storage 7410 Unified Storage System, an appliance cluster will support knowledge of up to two managed storage pools in addition to the system pool. A passive node will preload the configuration of a managed pool and then automatically import it on a fail-over event.

RAS Features

Beyond clustering capabilities, Sun Storage 7000 Management software provides a large collection of RAS features to help ensure that data remains available, including:

- *Automated restart of all software services on the system* — Based on the Service Management Facility, OpenSolaris daemons and the daemons associated with the software itself are automatically restarted upon software failure.
- *Offlining of hardware faults* — Diagnosis and offlining of hardware faults, including CPUs, memory, I/O adapters, and disk devices. Based on the Solaris Fault Management Architecture, this capability also provides a real-time error telemetry feed for use by Sun Service in root-cause analysis.
- *System software snapshot and rollback* — Whenever the appliance is upgraded, a set of rolling snapshots of previous software installs is maintained. These snapshots give administrators the ability to immediately roll the system back to a previous installation, if desired.

- *Hardware integration between FMA diagnosis and hardware indicators* — All of the Sun servers employed as appliances contain LEDs to indicate faulty components, aiding serviceability. FMA integration and platform specialization software light these indicators as FMA diagnoses faulty components.

Alerts and Watchlists

Sun Storage 7000 Management software also provides mechanisms for generating alerts corresponding to fault-management diagnosis results and other aberrant behaviors that can be detected on the system. Examples include power supply or fan failures, configuration errors, and network outages that are affecting the appliance. Alerts appear on the Web dashboard, and are saved persistently. Alerts can also be transmitted using other transports, including e-mail and SNMP traps.

The software also provides a set of built-in e-mail alerts associated with performance monitoring and capacity planning. Watchlists can be designed to send an e-mail when a system condition crosses a user-configurable threshold. Other alerts are generated automatically on a periodic basis, and either sent over e-mail or viewed through the administrative GUI.

Phone Home

Phone-home support is also provided by Sun Storage 7000 Management software. This capability provides an enterprise-class service experience suitable to Sun appliances. Phone home is supported through two transports, HTTPS and SMTP (encrypted e-mail). Phone-home support makes use of a unique Appliance Serial Number (ASN) issued by Sun at the time the appliance leaves the manufacturing facility, so that phone-home information can be immediately correlated with a given customer account and level of service entitlement.

Phone-home events are generated whenever a configuration change occurs, to query for software updates that may be available, and whenever a fault-management diagnosis event occurs.

IPMI and SNMP

All of the Sun servers used in Sun Storage 7000 Unified Storage Systems include an integrated service processor with an intelligent platform management interface (IPMI) stack for network management. Each server's service processor includes a dedicated network device that permits remote management of the system through IPMI. Key features offered through IPMI include the ability to power-cycle or reboot the system remotely, to retrieve hardware failure notification through the System Event Log (SEL) persisted on the service processor, and the ability to debug problems where the operating system is unable to boot.

The software also provides SNMP v2 and v3 MIB browsing as well as trap support for integrating Sun appliances with heterogeneous management tools. The software includes the standard enterprise MIBs for naming and locating the appliance, standard networking MIBs, and a fault management MIB. The appliances generate traps for all FMA events as problems are diagnosed on the system. SNMP visibility is also provided for storage consumption.

Chapter 5

Conclusion

As organizations strive to manage their rapidly-expanding storage infrastructure, they face challenges that are technical, managerial, and economic. Traditional NAS appliances have fallen short due to the proprietary nature of their software and the special-purpose nature of underlying hardware. Sun Storage 7000 Unified Storage Systems represent an innovative new approach that lets organizations use scalable, high-performance unified storage for more of their most important applications.

Sun Storage 7000 Unified Storage Systems are based on robust and balanced industry-standard Sun servers, removing performance bottlenecks, and offering a scalable and expandable product line. Innovative enterprise SSDs bring a new dimension, effective expanding caching capabilities and performance while keeping costs in line. Seamless Sun Storage 7000 Management software presents an easy-to-use administrative interface that requires little or no training to operate.

By using open technologies, Sun Storage 7000 Unified Storage Systems avoid the constraints, pitfalls, and costs of traditional storage appliances. OpenSolaris technology delivers performance and an active community for improvement. Solaris ZFS provide Hybrid Storage Pools that effectively combine system memory, enterprise SSDs, and conventional disk drives. DTrace Analytics gives administrators real-time access to in-depth business analytics, letting them respond and anticipate issues. Together, these technologies promise to change the very nature of unified storage.

For More Information

To learn more about Sun Storage 7000 Unified Storage systems products, please contact a Sun sales representative, or consult the related documents and Web sites listed in Table 4.

Table 4. Related Websites

Web Site URL	Description
sun.com/unifiedstorage	Product information
sun.com/blueprints/storage/7000	Technical best practices

